COMP102: DATA STRUCTURES WITH PYTHON

Stacks & Queues

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March 27, 2026

Plan

1. Learning Outcomes
2. What are abstract data types?
3. Stacks
4. Queues
5. Python Choices and Complexity
6. Encapsulating restricted access
7. Pitfalls & Takeaways

Learning Outcomes

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The point of the lecture is not to memorize method names. The point is to understand **restricted access** as a design choice.

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Key idea

An ADT describes **what** behavior we want, *not* **how** it is represented in memory.

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ADT (specification)

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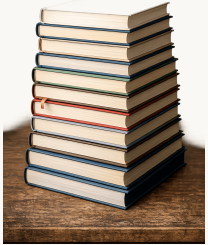
A good way to say it: **an ADT is a contract**; **a data structure is a way to satisfy that contract**.

Stacks

Intuition

Cds on a spindle.

Pile of dishes.



Pile of books.

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- `push(x)`: add `x` on top
- `pop()`: remove and return the top item

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Typical examples

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- function calls

Tracing a stack step by step

Start with an empty stack and apply the following operations:

`push(4), push(9), push(2), pop(), push(7), peek()`

Step	**Stack content (bottom → top)**	**Returned value**
start	[]	—
`push(4)`		

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`push(2)`		

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`pop()`		

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`push(9)`	`[4, 9]`	—
`push(2)`	`[4, 9, 2]`	—
`pop()`	`[4, 9]`	2
`push(7)`		

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`peek()`	`[4, 9, 7]`	7

For a stack, the next element to leave is always the current **top**.

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Direct use of `list`

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stack = []

stack.append(4)   # push(4)
stack.append(9)   # push(9)
stack.append(2)   # push(2)

x = stack.pop()   # x = 2

top = stack[-1]   # peek -> 9
empty = len(stack) == 0
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Example 3.4. Why this is a good fit

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Example 3.5. Why this is a good fit

`append()` and `pop()` at the end of a list are efficient operations, which matches the one-end discipline of a stack.

Where stacks appear

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Real systems

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- browser back / undo history

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A stack is often the right tool when the problem depends on the **most recent unfinished context**.

Queues

Intuition

Waiting in line at service counter.

Industrial conveyor belt and processing station.

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Mental model

Front Rear

4	9	2

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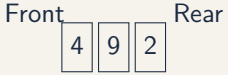

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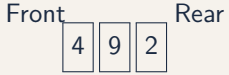

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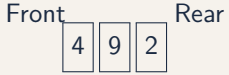

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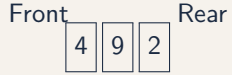

Typical examples

- waiting line
- printer jobs
- Task scheduling

Tracing a queue step by step

Start with an empty queue and apply the following operations:

`enqueue(4), enqueue(9), enqueue(2), dequeue(), enqueue(7), front()`

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start	[]	—
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`enqueue(2)`	`[4, 9, 2]`	—
`dequeue()`	`[9, 2]`	4
`enqueue(7)`		

Tracing a queue step by step

Start with an empty queue and apply the following operations:

`enqueue(4), enqueue(9), enqueue(2), dequeue(), enqueue(7), front()`

Step	**Queue content (front → rear)**	**Returned value**
start	`[]`	—
`enqueue(4)`	`[4]`	—
`enqueue(9)`	`[4, 9]`	—
`enqueue(2)`	`[4, 9, 2]`	—
`dequeue()`	`[9, 2]`	4
`enqueue(7)`	`[9, 2, 7]`	—
`front()`		

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`enqueue(9)`	`[4, 9]`	—
`enqueue(2)`	`[4, 9, 2]`	—
`dequeue()`	`[9, 2]`	4
`enqueue(7)`	`[9, 2, 7]`	—
`front()`	`[9, 2, 7]`	9

For a queue, the next element to leave is always the one that has been waiting the longest.

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Implementing a queue in Python

For queues, the standard Python tool is usually `collections.deque`.

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Implementing a queue in Python

Recommended approach

```python
from collections import deque

queue = deque()

queue.append(4)       # enqueue(4)
queue.append(9)       # enqueue(9)
queue.append(2)       # enqueue(2)

x = queue.popleft()   # x = 4

front = queue[0]      # peek/front -> 9
empty = len(queue) == 0
```

Implementing a queue in Python

Recommended approach

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from collections import deque

queue = deque()

queue.append(4)       # enqueue(4)
queue.append(9)       # enqueue(9)
queue.append(2)       # enqueue(2)

x = queue.popleft()   # x = 4

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empty = len(queue) == 0
```

Example 4.4. Why deque?

Implementing a queue in Python

Recommended approach

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from collections import deque

queue = deque()

queue.append(4)       # enqueue(4)
queue.append(9)       # enqueue(9)
queue.append(2)       # enqueue(2)

x = queue.popleft()   # x = 4

front = queue[0]      # peek/front -> 9
empty = len(queue) == 0
```

Example 4.5. Why deque?

A queue needs efficient operations at opposite ends. That is exactly what deque is designed for.

Why `list.pop(0)` is a bad habit for queues

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Naive version

```
queue = []
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queue.append(2)

x = queue.pop(0)    # works, but not a good implementation choice
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- The code is **behaviorally correct**: it does remove the oldest item.
- But removing index 0 shifts the remaining elements.
- So a solution can be correct and still be the wrong implementation choice.

Python Choices and Complexity

Choosing the right implementation

ADT	Operation	Good Python tool	Intuition	Complexity
Stack	push	`list.append()`	efficient at the end	O(1) amortized
Stack	pop	`list.pop()`	efficient at the end	O(1) amortized
Queue	enqueue	`deque.append()`	efficient at the rear	O(1) amortized
Queue	dequeue	`deque.popleft()`	efficient at the front	O(1) amortized
Queue (naive)	dequeue	`list.pop(0)`	shifting cost	O(n)

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Queue (naive)	dequeue	`list.pop(0)`	shifting cost	O(n)

The ADT tells us **what** behavior we want. Complexity helps us choose **how** to implement it.

Stack vs queue: side-by-side recap

Feature	**Stack**	**Queue**
Discipline	LIFO	FIFO
Insert operation	`push`	`enqueue`
Remove operation	`pop`	`dequeue`
Where items leave	top	front
Typical Python tool	`list`	`deque`
Common uses	undo, DFS, parsing	scheduling, BFS, buffering

Encapsulating restricted access

Why wrap stacks and queues in classes?

A class helps enforce restricted access: users interact with the *ADT operations*, not with the raw container.

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Problem with the direct approach

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When we write `stack = []` or `queue = deque()`, the underlying container is fully exposed. Nothing stops us from doing operations that violate the intended discipline.

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- `push`, `pop`, `peek`
- `enqueue`, `dequeue`, `front`
- `is_empty`

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Template: MyStack

Class skeleton

This template defines the interface of a stack without implementing it yet.

```python
class MyStack:
    def __init__(self):
        pass

    def push(self, value):
        pass

    def pop(self):
        pass

    def peek(self):
        pass

    def is_empty(self):
        pass
```

Template: `MyQueue`

Class skeleton

This template defines the interface of a queue without implementing it yet.

```python
class MyQueue:
    def __init__(self):
        pass

    def enqueue(self, value):
        pass

    def dequeue(self):
        pass

    def front(self):
        pass

    def is_empty(self):
        pass
```

Pitfalls & Takeaways

Common mistakes

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- Treating *behavior* and *implementation* as if they were the same idea.

A data structure is not just a place to store values. It is a promise about which operations are natural, efficient, and meaningful.

Takeaways

These structures are simple, but they are foundational. They reappear inside parsers, search algorithms, schedulers, interpreters, and many programming problems.

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